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MFEM/MGIS, a HPC mini-application targeting nonlinear thermo-mechanical simulations of nuclear fuels at mesoscale

Thomas Helfer [1], **Guillaume Latu** [1], **Raphaël Prat** [1], **Maxence Wangermez** [1], **and Francesca Cuteri** [1]

1 CEA, DES, IRESNE, DEC, Cadarache F 13108 St Paul Lez Durance

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Summary

The MFEM/MGIS application aims at efficiently use supercomputers in order to describe coupled multiphysics phenomena with a particular focus on thermo-mechanics.

This open-source library is based on several components as prerequisites: the MFEM (Modular Finite Element Methods) (Anderson et al., 2021; Kolev & Dobrev, 2010) library, the MGIS (MFront Generic Interface Support) (Helfer et al., 2020) library and the MFront code generator (Helfer et al., 2015).

Thanks to the features embedded within MGIS and MFront and thanks to specific developments, MFEM/MGIS adds several mechanical features compared to a pure MFEM approach. The library tackles some peculiarities of nonlinear mechanics. In particular, the support of complex constitutive laws and the management of advanced boundary conditions.

It provides a high level of abstraction based on a declarative text-based Application Programming Interface suitable for engineers.

MFEM-based Thermo-Mechanical Solver for Nuclear Fuel Simulations

MFEM, is a finite element library designed for current supercomputers but also for the upcoming exascale supercomputers. It provides many useful features for carrying out realistic simulations: support for curvilinear meshes, high order approximation spaces and different families of finite elements, interfaces to several types of parallel solvers (including matrix-free ones), preconditioners, and native support for adaptive non-conforming mesh refinement (AMR).

Originating from the applied mathematics and parallel computing communities, MFEM offers both performance and a large panel of advanced mathematical features. In particular, one can easily switch from one linear solver to another (direct or iterative), which is essential for the targeted application (Bernaud et al., 2024): microstructure and mesoscale modeling for nuclear fuel.

Statement of need

The solid mechanic applications in MFEM are mostly limited to simple constitutive equations such as elasticity and hyperelasticity, which is insufficient to address complex nuclear fuel simulations.

The aim of MFEM/MGIS project is to combine MFEM with the MFrontGenericInterfaceSupport (MGIS) project, an open-source C++ library handles all the kinds of behavior supported by the open-source MFront code generator.

In the field of nonlinear mechanics, this encompasses arbitrary complex behaviors that can describe damage, plasticity, viscoplasticity, phase change in both small or finite strain analyses. Generalized behaviors such as variational approaches to fracture, Cosserat media are supported by MFEM/MGIS. Strongly coupled thermo-mechanical behaviors and Cosserat media can be introduced by plugins and will be integrated in future versions of MFEM/MGIS.

Through its integration with MGIS and MFront, along with dedicated advancements, MFEM/MGIS has incorporated the subsequent mechanical attributes in contrast to the pure MFEM approach while maintaining parallel performance:

- Ability to handle several materials which distinct constitutive equations:
  - Micromorphique laws
  - Orthotropic laws
- Support for internal state variables (defined at integration points):
  - Use of MGIS data structures
  - Use of mechanical behavior laws generated by MFront
- Support for post-processing specific to nonlinear mechanic:
  - ComputeResultantForceOnBoundary: Compute the resultant of the inner forces on a boundary
  - ComputeMeanThermodynamicForcesValues: Compute the macroscopic stress and strain for each materials
  - ParaviewExportIntegrationPointResultsAtNodes: Paraview post processing files with internal state variables according to MGIS
- Flexible C++ API to easily define MFEM simulation features

These features are described in the following tutorial: https://thelfer.github.io/mfem-mgis/tutorial.html.

The implementation of high order meshes or finite elements is straight-forward.

The library tackles some peculiarities of nonlinear mechanics. In particular, the support of complex constitutive laws and the management of advanced boundary conditions.

The MFEM/MGIS library is written in C++17 language and provides a high level of abstraction based on a declarative text-based Application Programming Interface.

Several examples can be found on the open-source GitHub repository: https://github.com/latug0/mfem-mgis-examples, including simulation of the microstructural scale using a Representative Volume Element (RVE) of nuclear fuel.

Integrate MFEM/MGIS in an Open Source Ecosystem

The MFEM/MGIS library takes full advantage of an open-source software (OSS) stack. It benefits from the increasing maturity of several scientific tools that are combined.

Thus, within MFEM, one has many available choices to set the linear solver, such as: Hypre(Falgout & Yang, 2002), PETSc(Balay et al., 2023), MUMPS(Amestoy et al., 2001), SuperLU, UMFPACK(Davis, 2004) or other ones. Likewise, several preconditioners, partitioning libraries, or input mesh formats can be activated and used.

Combinations are highly configurable and almost all external libraries that relates to linear solvers are switchable.

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Software stack

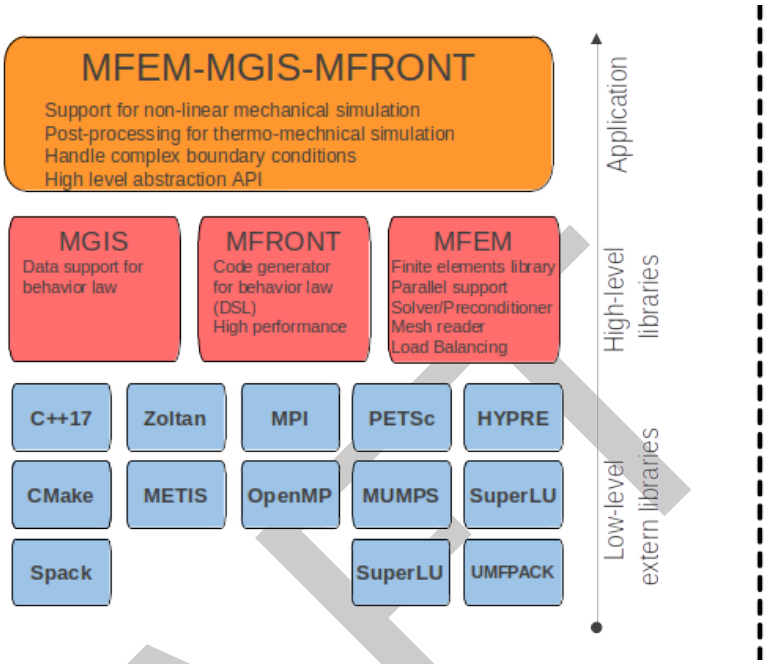

Figure 1: MFEM/MGIS Stack. Each library is open source.

The MFEM/MGIS software stack is presented on Figure 1. Most of low-level external libraries are required by `MFEM` and can be disabled. Hence, the minimal package requirements to build MFEM/MGIS on a HPC platform is typically:`C++17`, `MFEM`, `MGIS`, `TFEL`(MFront), CMake and `MPI`.

To handle the numerous accessible combinations, `Spack` (Gamblin et al., 2015) is really a cornerstone. This package manager simplifies building, installing, customizing, and sharing HPC software stacks. It provides a simple way for installing packages with cumbersome structures and lots of dependencies. `Spack` is an open-source package manager developed and maintained by community of HPC developers. Our setup that combines OSS allows for a fast and cheap access to advanced features embedded in the underlying libraries.

To complete the installation, you can enable the following options while compiling `MFEM`. The most interesting are the solver or preconditioner packages such as `Hypre`, `MUMPS`, and `PETSc` (including `SuperLU`, `UMFPACK`), and the load balancing packages `Zoltan` (Devine et al., 2000) and `Metis` (Karypis et al., 1997).

Numerical Results

Installation and deployment on desktop or large computers is shortened using the Spack package manager. Multi-material elastic modelling on computational clusters has been carried out with `MFEM/MGIS`. The observed scalability performance is good on a few thousands of CPU cores.

Despite the very high level of abstraction and the genericness of `MFEM/MGIS` (multi-material and arbitrary behaviors), the overhead appears reasonably limited. In the worst-case for `MFEM/MGIS`, an overhead of up to 30% has been observed compared to a pure `MFEM` version which provides very optimized and specialized kernels (the test was performed on a simulation with only two elastic materials).

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Several examples can be found on the open-source GitHub repository: https://github.com/latug0/mfem-mgis-examples.

Conclusion

This paper presents the `MFEM/MGIS` HPC application designed to address large scale thermo-mechanical simulation and recent supercomputers. Based on an open source software stack, it allows for the fine representation of microstructure in full 3D in the field of fuel modeling.

On the one hand, `MGIS` and `MFront` bring support for complexe nonlinear behaviours such as damage, plasticity, viscoplasticity capabilities. On the other hand, `MFEM` provides advanced finite elements schemes and parallel performance (tested on several thousands of cores until now). Open source approach was chosen mainly to promote collaboration, improve reproducibility, and reduce costs for development and maintenance.

Regarding performance portability on GPUs, MFEM already offers numerous algorithms such as partial assembly on GPUs, but MFEM/MGIS does not exploit these features yet. Work is underway to port behavior laws to the GPU (`MFront`) and associated data structures (`MGIS`). Note that other HPC libraries is considered for the future, such as `CUDA`, `RAJA` or `OCCA` for performance portability of MFEM/MGIS on GPU.

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